

VIRTUAL INTERNSHIP PRESENTATION

SQL & Database Management Systems

An 8-Week Industry Internship Program – EduSkills Academy

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Certificate ID: 2026-AFB5DB5F74 | Issued: July 16, 2026

What We'll Cover

- 1 About the internship program
- 2 Program snapshot at a glance
- 3 8-week learning journey
- 4 Skills and tools acquired
- 5 Capstone project
- 6 Learning outcomes & reflection
- 7 Conclusion

About the Internship Program

EduSkills Academy delivers structured, credentialed virtual internships that bridge academic learning and workplace readiness. The SQL & Database Management Systems track builds progressively — from relational fundamentals to production-grade, full-stack database skills.

1

Build a strong foundation in relational database design and SQL

2

Introduce NoSQL concepts through MongoDB

3

Develop data visualization skills using Power BI

4

Integrate databases with full-stack apps (Node.js & Python)

5

Cover performance, security, backups, and scalability

6

Deliver a capstone analytics dashboard project

Program Snapshot

8

Weeks of
Structured Learning

2

Full-Stack Frameworks
(Node.js & FastAPI)

1

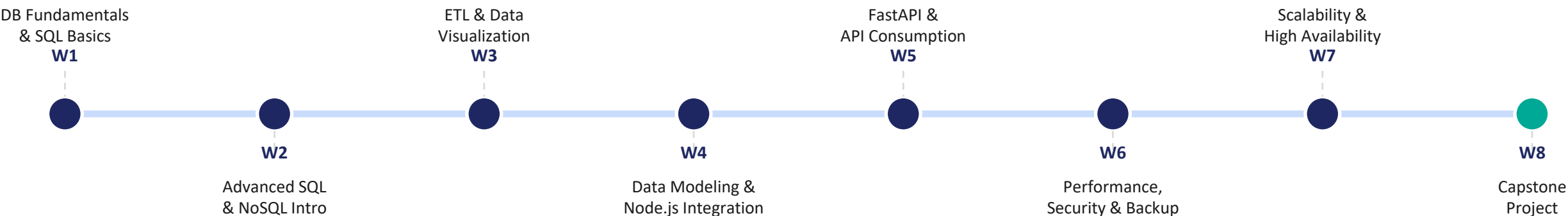
Capstone Analytics
Dashboard

SQL +
NoSQL

Database Paradigms
Covered

The 8-Week Learning Journey

From relational fundamentals to a full-stack, data-driven capstone



Weeks 1–2

Week 1 — Database Fundamentals & Relational Design

Build a foundation in relational database concepts and core SQL syntax.

- Course kick-off & program orientation
- Relational database design principles
- Normalization concepts (1NF, 2NF, 3NF)
- SQL basics: SELECT, WHERE, ORDER BY

Key outcome: *Could design a simple schema and write basic SQL queries.*

Week 2 — Advanced SQL & Intro to NoSQL

Extend SQL skills to multi-table operations and explore non-relational models.

- Advanced SQL: JOINS and subqueries
- Data aggregation (GROUP BY, HAVING)
- NoSQL concepts vs. relational systems
- MongoDB basics: documents & CRUD

Key outcome: *Could combine multi-table data and understood NoSQL use cases.*

Weeks 3–4

Week 3 — ETL Processes & Data Visualization

Learn how raw data becomes actionable insight via ETL and visualization.

- ETL concepts using SQL
- Cleaning & staging data for analysis
- Introduction to Power BI
- Building foundational dashboards

Key outcome: *Could move raw data into analysis-ready datasets and visualize it.*

Week 4 — Data Modeling & Node.js Integration

Connect database design to real application development (JS stack).

- Data modeling & relationships in Power BI
- Structuring DBs for reporting
- Integrating DBs with Node.js & Express
- Building basic API endpoints

Key outcome: *Connected a working database to a Node.js/Express backend.*

Weeks 5–6

Week 5 — FastAPI Integration & API Consumption

Apply integration principles using Python and consume external APIs.

- Integrating DBs with Python & FastAPI
- Designing RESTful endpoints
- Data integration from multiple sources
- Consuming third-party APIs

Key outcome: *Built database-backed API endpoints in FastAPI.*

Week 6 — Performance, Security & Backups

Understand how production databases are optimized and protected.

- Performance optimization & indexing
- Security fundamentals & access control
- Backup strategies & disaster recovery
- Common vulnerabilities & best practices

Key outcome: *Learned to keep databases fast, secure, and recoverable.*

Weeks 7–8

Week 7 — Scalability & High Availability

Study how databases scale and stay available under load.

- Vertical vs. horizontal scaling, sharding
- High availability & failover strategies
- Real-world case studies
- Industry best practices

Key outcome: *Understood how enterprise systems scale and stay available.*

Week 8 — Capstone & Advanced NoSQL

Apply all prior learning to a comprehensive capstone project.

- Advanced NoSQL querying (MongoDB)
- Designing the analytics dashboard
- Integrating SQL + NoSQL sources
- Final review & presentation

Key outcome: *Independently built a data-driven analytics dashboard.*

Skills and Tools Acquired

Relational Databases

SQL, schema design, normalization

NoSQL Databases

MongoDB, document modeling

Data Visualization

Power BI, data modeling

Backend Integration

Node.js, Express, Python, FastAPI

Data Engineering

ETL, data & API integration

Database Operations

Performance tuning, security, backup

Systems Design

Scalability, high availability

Project Delivery

End-to-end capstone execution

Capstone Project

Data-Driven Business Analytics Dashboard

Objective

Design and build an analytics dashboard that consolidates business data from both SQL and NoSQL sources into a single reporting layer supporting data-driven decisions.

Key Deliverables

- Normalized relational schema for core business data
- MongoDB structure for semi-structured data
- API endpoints via Node.js/Express or FastAPI
- Interactive Power BI dashboard of key metrics

1

Design SQL schema

2

Model NoSQL data

3

Build ETL pipeline

4

Expose via API

5

Visualize in Power BI

Note: Capstone scope expands on the certificate's project title with a representative technical approach.

Learning Outcomes & Reflection

Technical Growth

- Writing SQL from basic filters to multi-table joins
- Understanding relational vs. NoSQL trade-offs
- Connecting DBs to two backend ecosystems
- Awareness of performance, security & scalability

Professional Skills

- Self-directed, structured learning
- Independently planning a multi-stage project
- Translating business needs into technical solutions

Areas for Growth

- Advanced performance tuning at scale
- Cloud-managed database services
- More full-stack integration practice

Conclusion

This 8-week internship provided a comprehensive, hands-on path from relational fundamentals to a full data-driven analytics capstone — building skills in SQL, MongoDB, Power BI, Node.js, and FastAPI that form a strong foundation for roles in data engineering, backend development, or business intelligence.

Thank You

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